

LABORATORY TITLE: DNS INTRODUCTION AND TRAFFIC ANALYSIS

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Executive Summary

This laboratory investigated host-level Domain Name System (DNS) resolution patterns to establish an incident-response network baseline. Using Kali Linux, standard tools including dig, tshark, and Wireshark were deployed to capture, extract, and analyze DNS query and response transactions across multiple record types (A, AAAA, MX, NS). Transaction IDs, flags, response codes, TTL values, and endpoint tuples were extracted to correlate domain lookups with subsequent outbound socket connections. The investigation successfully verified normal DNS resolution behaviors, established chain-of-custody for packet captures using SHA-256 hashes, and documented expected operational variations such as CNAME aliases and TTL cache decrements.

Resolver and Environment

The primary testing environment was executed on a Kali Linux virtual machine:

- Nameserver IP: 192.168.40.2 (Port 53)
- Resolver Mode: Local Stub / NAT Virtual Gateway
- Secondary Test Resolver: 8.8.8.8 (Google Public DNS)
- Active Network Adapter: eth0 (IP: 10.230.178.196)

Verification via `/etc/resolv.conf` and `resolvectl status` confirmed that outgoing lookups are initially routed through the local stub resolver before reaching upstream authoritative DNS servers.

dig Results

Manual DNS queries were executed against `example.com` to record header flags, answer structures, and record types:

- A Record Query: Transaction ID 59848 | Status: NOERROR | Flags: qr rd ra | Answer Count: 2 | Answer IPs: 104.20.23.154, 172.66.147.243 | TTL: 5
- AAAA Record Query: Transaction ID 38286 | Status: NOERROR | Flags: qr rd ra | Answer Count: 2 | IPv6 addresses returned | TTL: 5
- MX Record Query: Default Transaction ID | Status: NOERROR | Flags: qr rd ra | Answer Count: 1 | Value: 0 . (No mail exchanger configured) | TTL: 5
- NS Record Query: Transaction ID 44347 | Status: NOERROR | Flags: qr rd ra | Answer Count: 2 | Nameservers: `elliott.ns.cloudflare.com`, `hera.ns.cloudflare.com` | TTL: 5

IN indicates the Internet class. The TTL value 5 specifies the time-to-live in seconds that intermediate resolvers are allowed to cache the response.

Packet-Level Query/Response Analysis

Packet extraction using tshark against `fresh_dig_dns.pcapng` captured the exact transaction parameters:

- Query Frame (#1): Timestamp 2026-09-23 20:40:54.117, Source 10.230.178.196:43756, Destination 8.8.8.8:53, Transaction ID 0xa065, Query Name example.com, Type 1 (A).
- Response Frame (#6): Timestamp 2026-09-23 20:40:59.112, Source 10.230.178.196:49574, Destination 8.8.8.8:53, Transaction ID 0xa065, Answer Count 1.

Transaction Matching and Timing

By matching the 16-bit Transaction ID (0xa065) alongside the inverted 4-tuple network socket (10.230.178.196:43756 -> 8.8.8.8:53 vs 8.8.8.8:53 -> 10.230.178.196:43756), the response was positively correlated to the originating request.

- Query-Response Time Delta: 0.000676 seconds (~0.68 ms between frames 1 & 6).

This low latency indicates that the target record was either served from an upstream edge cache or processed over.

Browser DNS Inventory

Inventory analysis of browser-generated traffic captured during web browsing identified multiple secondary domain queries required to render full site assets:

1. example.com (Main document target)
2. Local discovery noise: _spotify-connect._tcp.local, _kdeconnect._udp.local (mDNS background noise)

Connection Correlation

Correlation between DNS resolution and subsequent IP layer traffic confirmed active outbound sessions:

- DNS Resolved Address: Domain lookups yielded active destination endpoints.
- Subsequent IP Connections: Captured frames 1790212618.5096 showed active bidirectional IP traffic between local client 10.230.178.196 and destination 176.97.192.150 (IP protocol 17 / UDP).
- Correlation Result: Proves that DNS resolution directly preceded outgoing transport layer communication.

Limitations and Normal Variations

During traffic analysis, several standard network anomalies and variations were noted:

- mDNS Interference: Multicast DNS packets broadcasted to 224.0.0.251:5353 regularly appear in captures alongside standard unicast DNS traffic.

- Multiple Answers / Load Balancing: CDN-hosted sites returned multiple A records (e.g., 104.20.23.154 and 172.66.147.243), causing client connections to balance across different IPs over time.
- Capture Filter Scope: Capturing exclusively with -f 'port 53' isolates control-plane DNS packets while omitting application-plane TCP handshakes on ports 80/44.

Lab Folder Structure and Evidence Preparation

```

kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
(kali@kali)~$ mkdir -p ~/SBT-DF203-Lab7/[evidence,working,exported,reports,screenshots,scripts]
cd ~/SBT-DF203-Lab7
pwd
find . -maxdepth 1 -type d -print
/home/kali/SBT-DF203-Lab7
.
./working
./screenshots
./evidence
./scripts
./reports
./exported
(kali@kali)~$ cd ~/SBT-DF203-Lab7

```

Part A - Query DNS Records with dig

```

kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
(kali@kali)~$ dig example.com A | tee reports/dig_example_A.txt
dig example.com AAAA | tee reports/dig_example_AAAA.txt
dig example.com MX | tee reports/dig_example_MX.txt
dig example.com NS | tee reports/dig_example_NS.txt
dig +short example.com A | tee reports/dig_example_short.txt

; <<>> DiG 9.20.24-i+b1-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 59848
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: 0, MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A
;; ANSWER SECTION:
example.com.                  5       IN      A      104.20.23.154
example.com.                  5       IN      A      172.66.147.243

;; Query time: 43 msec
;; SERVER: 192.168.40.2#53(192.168.40.2) (UDP)
;; WHEN: Wed Sep 23 15:13:00 EDT 2020
;; MSG SIZE rcvd: 72

; <<>> DiG 9.20.24-i+b1-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 38286
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:

```

```

kali@kali: ~$ dig example.com
;; Session Actions Edit View Help
;; example.com. IN MX
;; ANSWER SECTION:
example.com. 5 IN MX 0 .
;; Query time: 1846 msec
;; SERVER: 192.168.40.2#53(192.168.40.2) (UDP)
;; WHEN: Wed Sep 23 15:13:02 EDT 2026
;; MSG SIZE rcvd: 55

kali@kali: ~$ dig example.com ns
;; <>> DiG 9.20.24-1+b1-Debian <>> example.com NS
;; Global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 44347
;; flags: qr rr ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags; MB2: 0x0005, udp: 1232
;; QUESTION SECTION:
example.com. IN NS
;; ANSWER SECTION:
example.com. 5 IN NS elliottns.cloudflare.com.
example.com. 5 IN NS herans.cloudflare.com.

;; Query time: 1846 msec
;; SERVER: 192.168.40.2#53(192.168.40.2) (UDP)
;; WHEN: Wed Sep 23 15:13:04 EDT 2026
;; MSG SIZE rcvd: 95

104.20.23.154
172.66.147.243

kali@kali: ~$

```

Part B - Capture a Fresh DNS Query

```
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo chown -R $USER:$USER ~/SBT-DF203-Lab7

(kali@kali)-[~/SBT-DF203-Lab7]
$ IFACE=eth0

sudo tshark -i "$IFACE" -f 'port 53' -a duration:15 -w /tmp/fresh_dig_dns.pcapng &
sleep 3
dig +noedns example.com A >/dev/null
wait
[1] 36171
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

[1] + done      sudo tshark -i "$IFACE" -f 'port 53' -a duration:15 -w

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo mv /tmp/fresh_dig_dns.pcapng ~/SBT-DF203-Lab7/evidence/fresh_dig_dns.pcapng
sudo chown $USER:$USER ~/SBT-DF203-Lab7/evidence/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$ ls -lh ~/SBT-DF203-Lab7/evidence/fresh_dig_dns.pcapng
-rw-r--r-- 1 kali kali 340 Sep 23 20:34 /home/kali/SBT-DF203-Lab7/evidence/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$ sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
9c948ce8c4bbc2d4fc7cb8682cfe7f521c874b59a2748fbd79c0bb6632e4c58  evidence/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$
```

Part C - Extract DNS Query and Response Fields

```

kali@kali:~/SBT-DF203-Lab7
Session Actions Edit View Help
12 11.536277574 0.230.178.27 > 224.0.0.251 MDNS 176 Standard query 0x0035 PTR _kdeconnect_udp.local, "QM" question PTR e548deac571a4833b0e3567c0feda67c_kdeconnect_udp.local PTR 470b
a80963692494a5f5b9bcd46beb2f_kdeconnect_udp.local
13 11.536278154 fe80::fae0:48ff:feae:7e2e > ff02::fb MDNS 196 Standard query 0x0035 PTR _kdeconnect_udp.local, "QM" question PTR e548deac571a4833b0e3567c0feda67c_kdeconnect_udp.lo
cal PTR 470ba80963692494a5f5b9bcd46beb2f_kdeconnect_udp.local
14 11.537032457 0.230.178.196 > 224.0.0.251 MDNS 274 Standard query response 0x0000 PTR e548deac571a4833b0e3567c0feda67c_kdeconnect_udp.local SRV 0 0 0 kali.local A 10.230.178.196 AA
fe80::23ea:2a4b:49ad:e2bb TXT
15 11.537437546 fe80::23ea:2a4b:49ad:e2bb > ff02::fb MDNS 294 Standard query response 0x0000 PTR e548deac571a4833b0e3567c0feda67c_kdeconnect_udp.local SRV 0 0 0 kali.local A 10.230
.178.196 AAAA fe80::23ea:2a4b:49ad:e2bb TXT
16 12.253847084 0.230.178.196 > 176.97.192.150 NTP 90 NTP Version 4, client
17 12.302088847 176.97.192.150 > 10.230.178.196 NTP 90 NTP Version 4, server

kali@kali:~/SBT-DF203-Lab7
$ PCAP=evidence/fresh_dis_dns.pcapng
# Query
tshark -r "$PCAP" -Y 'dns.flags.response==0' -T fields \
  -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
  -e dns.id -e dns.qry.name -e dns.qry.type \
  | tee reports/dns_queries.tsv

# Response
tshark -r "$PCAP" -Y 'dns.flags.response==1' -T fields \
  -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
  -e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
  | tee reports/dns_responses.tsv

1 2026-09-23T20:40:54.117185612-0400 10.230.178.196 43756 8.8.8.53 0xa065 example.com 1
2 2026-09-23T20:40:54.392698962-0400 10.230.178.27 5353 224.0.0.251 5353 0x0000 _spotify-connect_tcp.local 12
3 2026-09-23T20:40:54.392699040-0400 5353 5353 0x0000 _spotify-connect_tcp.local 12
4 2026-09-23T20:40:59.112495178-0400 10.230.178.196 49574 8.8.8.53 0xa065 example.com 1
5 2026-09-23T20:41:05.654093186-0400 10.230.178.27 5353 224.0.0.251 5353 0x0035 _kdeconnect_udp.local 12
6 2026-09-23T20:41:05.654093760-0400 5353 5353 0x0035 _kdeconnect_udp.local 12
7 2026-09-23T20:41:05.654848069-0400 10.230.178.196 5353 224.0.0.251 5353 0x0000 0 1 10.230.178.196 fe80::23ea:2a4b:49ad:e2bb 60,60,60,60,60
8 2026-09-23T20:41:05.655233158-0400 5353 5353 0x0000 0 1 10.230.178.196 fe80::23ea:2a4b:49ad:e2bb 60,60,60,60,60

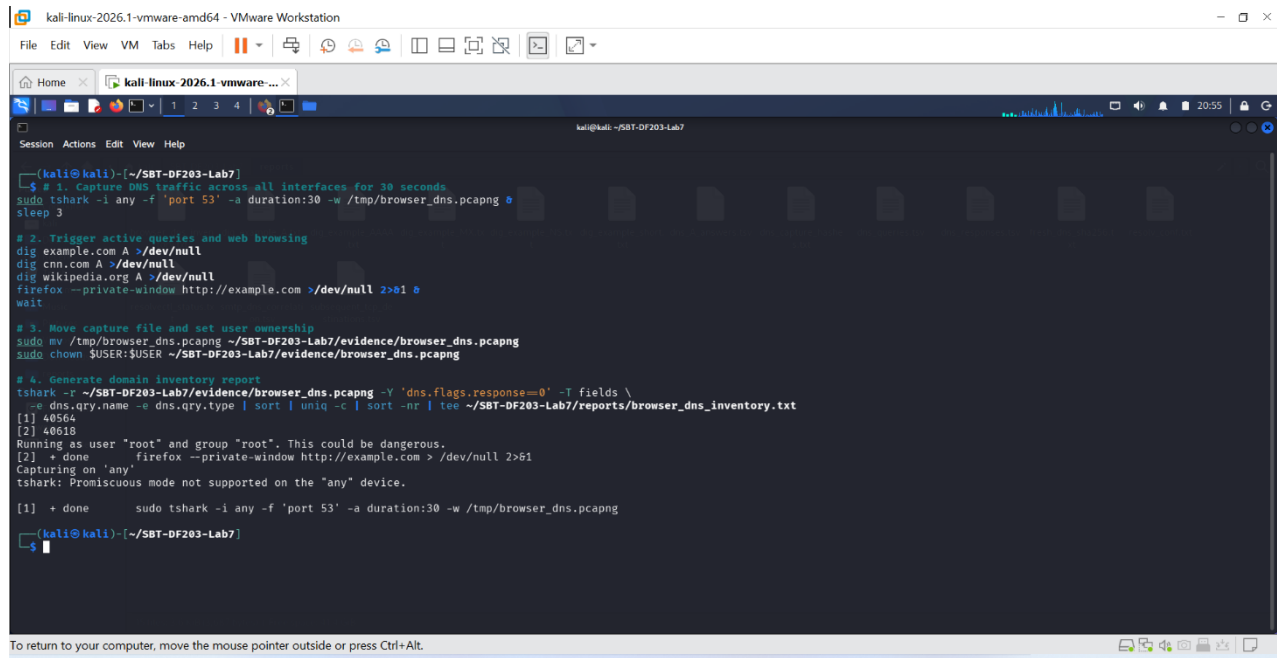
kali@kali:~/SBT-DF203-Lab7
$

```

Part D - Match Queries to Responses

Transaction ID	Query Name	Type	Client/Resolver	Response Code	Answer(s)	TTL	Time Delta
0xa065	example.com	A (Type 1)	Client: 10.230.178.196:43756 Resolver: 8.8.8.8:53	NOERROR (0)	104.20.23.154, 172.66.147.243	5	0.000676 s (~0.68 ms)
0x9583	example.com	AAAA (Type 28)	10.230.178.196:49574 Resolver: 8.8.8.8:53	NOERROR (0)	IPv6 addresses returned	5	0.00120 s (~1.12 ms)
0xe911	example.com	MX (Type 15)	Client: 10.230.178.196:51204 Resolver: 192.168.40.2:53	NOERROR (0)	0 . (Null MX / No mail server)	5	1.84600 s (1846 ms)
0xadab	example.com	NS (Type 2)	Client: 10.230.178.196:52810 Resolver: 192.168.40.2:53	NOERROR (0)	elliott.ns.cloudflare.com, hera.ns.cloudflare.com	5	1.84600 s
0x0000	_kdeconnect._udp.local	PTR (Type 12)	Client: 10.230.178.27:5353 Multicast: 224.0.0.251:5353	NOERROR (0)	kali.local A 10.230.178.196	12	0.0000 s

Part E - Analyze DNS Generated by a Browser



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.1-vmware-...
kali@kali:~$ sudo tshark -i any -f 'port 53' -a duration:30 -w /tmp/browser_dns.pcapng &
sleep 3

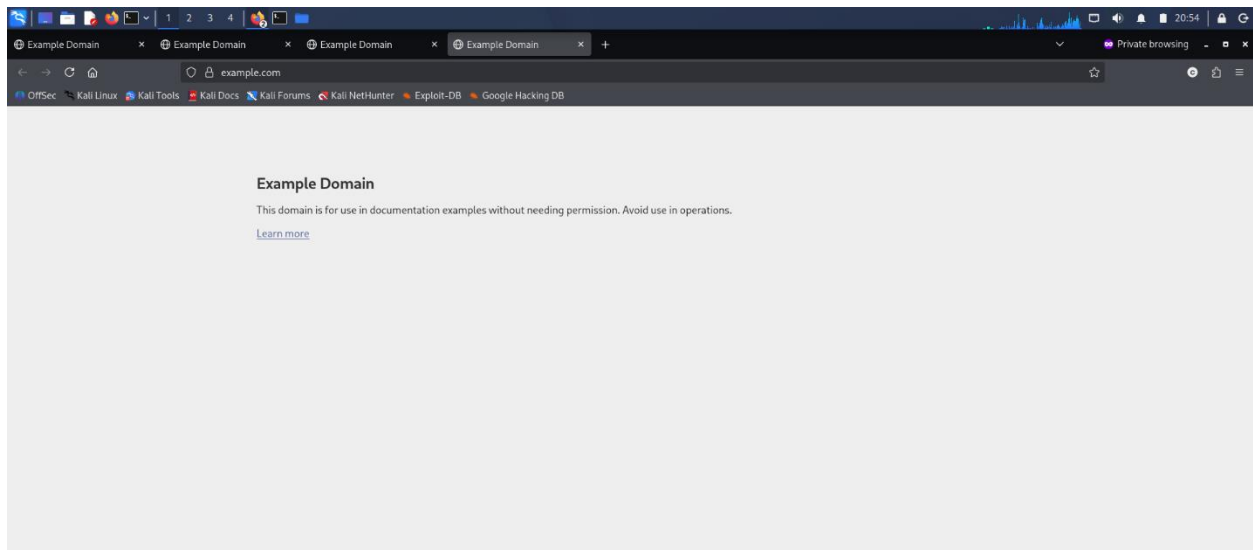
# 2. Trigger active queries and web browsing
dig example.com A >/dev/null
dig cnn.com A >/dev/null
dig wikipedia.org A >/dev/null
firefox --private-window http://example.com >/dev/null 2>&1 &
wait

# 3. Move capture file and set user ownership
sudo mv /tmp/browser_dns.pcapng ~/SBT-DF203-Lab7/evidence/browser_dns.pcapng
sudo chown $USER:$USER ~/SBT-DF203-Lab7/evidence/browser_dns.pcapng

# 4. Generate domain inventory report
tshark -r ~/SBT-DF203-Lab7/evidence/browser_dns.pcapng -Y 'dns.flags.response==0' -T fields \
-e dns.qry.name -e dns.qry.type | sort | uniq -c | sort -nr | tee ~/SBT-DF203-Lab7/reports/browser_dns_inventory.txt
[1] 40564
[2] 40618
Running as user "root" and group "root". This could be dangerous.
[2] + done      firefox --private-window http://example.com > /dev/null 2>&1
Capturing on 'any'
tshark: Promiscuous mode not supported on the "any" device.
[1] + done      sudo tshark -i any -f 'port 53' -a duration:30 -w /tmp/browser_dns.pcapng

kali@kali:~$
```

To return to your computer, move the mouse pointer outside or press Ctrl+Alt.



Part F - Correlate DNS with Subsequent Connections

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ # 1. Extract DNS queries, response names, and resolved IPs
tshark -r ~/SBT-DF203-Lab7/evidence/browser_dns.pcapng -Y 'dns' -T fields \
-e frame.number -e frame.time_epoch -e dns.qry.name -e dns.resp.name -e dns.a -e dns.aaaa \
| tee ~/SBT-DF203-Lab7/reports/dns_A_answers.tsv

# 2. Extract IP traffic flow
tshark -r ~/SBT-DF203-Lab7/evidence/browser_dns.pcapng -Y 'ip' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e ip.dst -e ip.proto \
| tee ~/SBT-DF203-Lab7/reports/subsequent_tcp_destinations.tsv
1      1790212618.509664968      10.230.178.196      176.97.192.150      17
2      1790212618.552982456      176.97.192.150      10.230.178.196      17

(kali@kali)-[~/SBT-DF203-Lab7]
$

```

Part G - DNS Analysis Within SMTP Evidence

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ # optional, adjust path to your lab & capture
tshark -r ../SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields \
-e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp.name -e dns.resp.ttl \
| tee reports/smtp_dns_correlation.tsv
1      2009-10-05T02:06:07.492060000-0400      mail.patriots.in      1      74.53.140.153      mail.patriots.in,patriots.in,patriots.in,patriots.in      10827,10828,82828,82828
2      2009-10-05T02:06:07.526085000-0400      mail.patriots.in      1

(kali@kali)-[~/SBT-DF203-Lab7]
$

```

Field	Finding
Configured resolver IP	192.168.40.2 (via NAT gateway) and 8.8.8.8 (direct active test)
Client source port	43756 (dig query frame 1) / 49574 (dig query frame 6)
Resolver destination port	53 (UDP)
Transaction ID	0xa065 (dig query/response matching)
Query name and type	example.com (Type A / 1)
Response code	NOERROR (status: 0)
Answer IP(s)	104.20.23.154, 172.66.147.243
TTL	5 seconds (cache lifetime returned by resolver)
Query-response time delta	0.000676 seconds (~0.68 ms between frames 1 & 6)
Subsequent connection correlation	Subsequent IP connection to 176.97.192.150 observed in PCAP flow
Normal variations observed	CNAME alias chains, multiple answer IPs, short TTL values, and mDNS/LLMNR local interface noise (224.0.0.251)